Fourier Transforms

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1 Introduction

Fourier analysis is the approximation of periodic functions by summing simpler, period trigonometric functions. The reason we want to do this is that trigonometric functions are simpler periodic functions to work with compared with constructing some new periodic function. Additionally, trigonometric functions also show up in the natural world, from representing sound waves to heat distribution. By representing these functions mathematically, we can create easier methods to perform certain calculations and operations to manipulate these periodic functions to get what we want.

2 Fourier Series

We define a Fourier series as follows. For a period function $f(x)$, its Fourier series is defined as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

with

$$a_k = \frac{2}{\pi}\int_0^{\pi} f(x)\cos kx\,dx \text{ and } b_k = \frac{2}{\pi}\int_0^{\pi} f(x)\sin kx\,dx,$$

and a_0 is some constant term.

To find the coefficients of any Fourier series, we want to isolate each of a_k, b_k, so we get some expression in terms of $f(x)$. To do this, we can consider multiplying by either $\sin mx$ or $\cos mx$ and taking the integral from 0 to π because we know that

$$\begin{aligned}
&\int_0^{\pi} \sin nx \sin mx\,dx = 0 \text{ if } m \neq n,\\
&\int_0^{\pi} \cos nx \cos mx\,dx = 0 \text{ if } m \neq n, \text{ and}\\
&\int_0^{\pi} \sin nx \cos mx\,dx = 0.
\end{aligned}$$

However, this may seem somewhat unmotivated if we were just multiplying and taking the integral. One alternative thought process is that the sines and cosines can be represented as vectors, and we project $f(x)$ onto the trigonometric functions. What's nice about vectors is that they lead us to orthogonality, and when two vectors are orthogonal, their dot product is 0. Now we just need to consider how to take the dot product of a function.

2.1 Linear Motivation

Consider two functions $f(x)$ and $g(x)$. Then, we take the two vectors with infinite dimensions

$$f : \langle f(x), f(x+\Delta x), f(x+2\Delta x), \cdots \rangle \text{ and } g : \langle g(x), g(x+\Delta x), g(x+2\Delta x), \cdots \rangle$$

with Δx approaching 0.

Note that when we dot f and g, the components of f and g multiply with each other and add to get the final value. This is analogous to finding the area under the curve $h(x) = f(x)g(x)$, so

$$f \cdot g = \int h(x)\, dx = \int f(x)g(x)\, dx.$$

We can show that $\sin nx$ and $\sin mx$ are orthogonal to each other since the integral of their products are equal to 0, and similarly for $\cos nx$ and $\cos mx$, and $\sin nx$ and $\cos mx$. Assuming we know that these functions are all orthogonal to each other, multiplying by the trig functions and taking the integral seems much less random now.

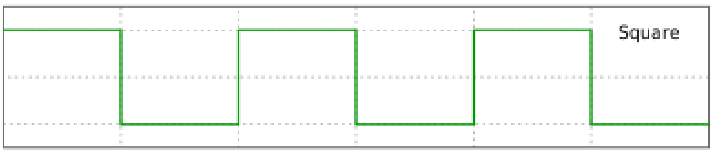
3 Square Function Example

To get a better understanding of Fourier series, let's try approximate the piece wise square function $F(x)$, with one period defined as

$$F(x) = \begin{cases} 1 & \text{if } 0 \le x < \pi \\ -1 & \text{if } \pi \le x < 2\pi \end{cases}$$

from 0 to 2π using Fourier series.

Firstly, let's graph a section of this function to get a sense of what we're working with.



Since our function is odd, we only need to consider the sine portion of the Fourier series. Then,

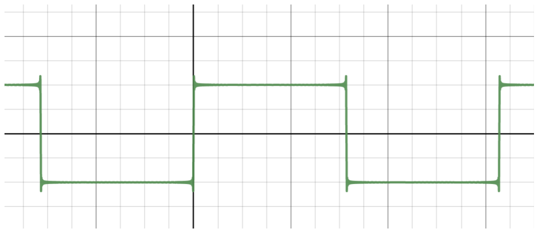
$$b_n = \frac{2}{\pi}\int_0^{\pi} \sin(nx)\, dx = \frac{2}{\pi}\left[\frac{-\cos nx}{n}\right]_0^{\pi} = \frac{-2\cos \pi n}{\pi n}.$$

Note that all even $b_n = 2k$ will become 0 since $\cos 2kx = 0$ for all $k \in \mathbb{N}$.

Since our function is centered at 0, $a_0 = 0$ and we don't have to include it in our expression. Therefore, we can write our series as

$$F(x) = \sum_{n=1}^{\infty} \frac{2 - 2\cos \pi n}{\pi n} \sin nx.$$

As a reality check, we can graph this on Desmos and sure enough, we get a graph very similar to the square function above.



4 Exponential Fourier Series

4.1 Functions with periods of 1

Recall that for any periodic function $f(x)$, we can write it as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx).$$

Note that the period of this function is 2π. For convenience, let's let the period of $f(x)$ instead be 1 so that we can easily change it in the future (this is especially useful for when we take the transform). Then, if we let $0 \leq t \leq 1$, we can rewrite $f(x)$ in terms of t.

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi nt) + b_n \sin(2\pi nt)$$

The sine and cosine sum also reminds us of something else. Recall Euler's Formula. For all real θ,

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

One can prove this by relating the power-series expansions of e^x, the sine function, and the cosine function. More specifically,

$$\begin{aligned}
e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \cdots \\
&= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \cdots \\
&= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!}\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!}\right) \\
&= \cos\theta + i\sin\theta.
\end{aligned}$$

Thus, we get our identity.

One thing to note is that this proof relies on the fact that imaginary numbers act the same way as real numbers do when exponential. However, this can be proven, but for the sake of simplicity, we will take this for granted. This also gives some interesting properties about sine and cosine functions.

For all real θ, we can write $\cos\theta$ as

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

and we can write $\sin\theta$ as

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Now, we can rewrite $f(t)$ in terms of e.

$$\begin{aligned}
f(t) &= a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi nt) + b_n \sin(2\pi nt) \\
&= a_0 + \sum_{n=1}^{\infty} a_n \frac{e^{2\pi int} + e^{-2\pi int}}{2} + b_n \frac{e^{2\pi int} - e^{-2\pi int}}{2i} \\
&= a_0 + \sum_{n=1}^{\infty} \frac{a_n}{2} e^{2\pi int} + \frac{a_n}{2} e^{-2\pi int} + \frac{b_n}{2} e^{2\pi int} - \frac{b_n}{2i} e^{-2\pi int} \\
&= a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{b_n}{2} i\right) e^{2\pi int} + \left(\frac{a_n}{2} + \frac{b_n}{2} i\right) e^{-2\pi int}
\end{aligned}$$

From here, we notice that we can let $c_n = \frac{a_n}{2} - \frac{b_n}{2} i$ and $c_{-n} = \frac{a_n}{2} + \frac{b_n}{2} i$. Additionally, notice that if we let $n = 0$ and sum the two coefficients together, $2c_0 = a_0$. In fact, textbooks generally have that the constant term of a trigonometric Fourier series as $\frac{a_0}{2}$, but for the sake of clarity, we let the constant term be a_0. However, now we can completely ignore the constant term, and instead in our Fourier series write $f(t)$ as

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi int}.$$

4.2 Motivation for coefficients

Rather than the linear motivation for finding the coefficients of a Fourier series by showing that two trigonometric functions are orthogonal, we will find an alternative motivation for multiplying by a function and taking the integral.

Suppose we wish to find the value of c_k. We are given

$$f(t) = \cdots + c_1 e^{2\pi it} + c_2 e^{4\pi it} + \cdots + c_k e^{2\pi ikt} + \cdots .$$

A common first step would be to isolate the c_k term by multiplying by $e^{-2\pi ikt}$. This gives us

$$f(t)e^{-2\pi ikt} = c_k + \sum_{n=-\infty, n\neq k}^{\infty} c_n e^{2\pi i(n-k)t}.$$

Now we wish to apply some operation such that all the other c_n terms become 0. Recall that $f(t)$ is periodic with period 1, and so is $e^{2\pi i(n-k)t}$. Additionally, e has a nice integral, so we may try integrating both sides from 0 to 1 with respect to t. For some generic n with $n \neq k$,

$$\begin{aligned}
\int_0^1 c_n e^{2\pi i(n-k)t}\,dt &= c_n \int_0^1 e^{2\pi i(n-k)t}\,dt \\
&= c_n \left[\frac{1}{2\pi in - kt} e^{2\pi i(n-k)t}\right]_0^1 \\
&= c_n \left(e^{2\pi i(n-k)} - e^0\right) \\
&= c_n(1-1) \\
&= 0.
\end{aligned}$$

With $\int_0^1 c_k\,dt$, we simply get c_k. Thus,

$$c_k = \int_0^1 f(t)e^{-2\pi ikt}\,dt.$$

For the sake of notation, we define $\hat{f}(n) = c_n$ for all valid n. The Fourier coefficient the nth term in the Fourier series of $f(t)$ is

$$\hat{f}(n) = \int_0^1 f(t)e^{-2\pi int}\,dt.$$

One can show that regardless of the start and end of the interval used for integration, so long as the interval is of length 1, the Fourier coefficient still holds true. More specifically,

$$\hat{f}(n) = \int_h^{h+1} f(t)e^{-2\pi int}\,dt$$

for $h \in \mathbb{R}$. A commonly used form is

$$\hat{f}(n) = \int_{-1/2}^{1/2} f(t)e^{-2\pi int}\,dt.$$

4.3 Functions with other periods

Let's say some periodic function $f(t)$ has a period of T. We can let $g(t) = f(Tt)$ so that $g(t)$ has period 1. If we let $s = Tt$, then

$$f(s) = g(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi int} = \sum_{n=-\infty}^{\infty} c_n e^{2\pi ins/T}.$$

We can apply a similar method as the previous section to find that

$$\hat{f}(n) = \frac{1}{T}\int_{-T/2}^{T/2} f(s)e^{-2\pi ins/T}\,ds.$$

Note that this is essentially just the original Fourier coefficient except scaled down by the period T. We can now replace s with t, and our new Fourier series is essentially the same as before.

Now that we have a more generalized form of the Fourier series, let's generalize it even further and consider when the period of the function is infinity.

5 Fourier Transform

The Fourier transform takes in any function, regardless if its periodic or not, and still expresses it as the sum of sine and cosine functions. To start, let's first just consider taking the limit of $\hat{f}(n)$ as T goes to infinity.

$$\lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t)e^{-2\pi int/T}\, dt$$

However, notice that

$$\left| \int_{-T/2}^{T/2} f(t)e^{-2\pi int/T}\, dt \right| \leq \int_{-T/2}^{T/2} \left| f(t) \right| \left| e^{-2\pi int/T} \right| dt = \int_{-T/2}^{T/2} \left| f(t) \right| dt,$$

which is just an expression independent of n and T. As a result, the above limit will tend towards 0, which is not what we want. Therefore, we scale everything up by T. Additionally, note that n/T is discrete, as $n \in \mathbb{N}$. We let $s = n/T$ to include all real terms, and thus we get

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(t)e^{-2\pi ist}\, dt,$$

and we're done. Hence, the Fourier transform of a function $f(t)$ is defined to be

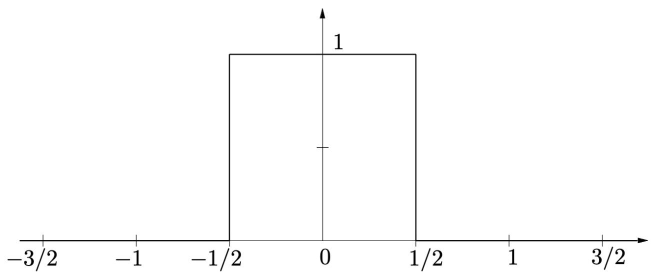
$$\hat{f}(s) = \int_{-\infty}^{\infty} f(t)e^{-2\pi ist}\, dt.$$

Here's an example of how we can use the Fourier transform to express non periodic functions in terms of periodic functions.

The top hat function is defined as

$$\Pi(t) = \begin{cases} 1 & \text{if } |t| < 1/2 \\ 0 & \text{if } |t| \geq 1/2 \end{cases}$$

Perform the Fourier transform on $\Pi(t)$. As a hint, here is the graph of the top hat function.



If all goes well, the answer should be

$$\hat{\Pi}(s) = \frac{\sin \pi s}{\pi s}.$$